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“loosely consolidated (rubble-pile) asteroid” (13). Itokawa is the first S-type asteroid showing such high porosity, and the first subkilometer-sized small asteroid showing a rubble-pile structure rather than a solid monolithic structure.

Two distinct types of terrain are found on Itokawa, rough terrain with numerous boulders and smooth terrain covered with regolith layer (6), which suggests a complex history but does not predicate a heterogeneous composition. In addition, no clear regional difference in the normalized x-ray intensity ratios of Mg/Si and Al/Si has been found on the surface of asteroid Itokawa (14), indicating homogeneous composition. NIRS reported olivine-rich mineral assemblages similar to LL5 and LL6 chondrites, with variations in albedo and absorption band depth more than 10%, but this diversity may be consistent with differences in freshness and/or particle size (11), and it does not suggest the presence of unusual inhomogeneous materials.

As noted in (15, 16), the size of the asteroid, 150 to 1000 m in radius, may be a transition size between monolithic structure and rubble-pile structure. The large porosity of ~40% and the roughness of the surface found on Itokawa strongly

suggest a rubble-pile structure. The 12.1324-hour spin period of Itokawa (1) is far above the critical value of 2 hours, where a rubble-pile structure cannot withstand centrifugal forces [e.g., (17)], and is also consistent with a rubble-pile structure.

The internal structure of the asteroid Itokawa gives us a hint of its origin. It is predicted, on the basis of the numerical simulation of orbital evolution of asteroid Itokawa, that the most probable source region of asteroid Itokawa is the inner part of the main belt (18). High porosity in asteroid Itokawa may be the result of gravitational aggregation of the collision fragments.

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REPORT

Pole and Global Shape of 25143 Itokawa

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The locations of the pole and rotation axis of asteroid 25143 Itokawa were derived from Asteroid Multiband Imaging Camera data on the Hayabusa spacecraft. The retrograde pole orientation had a right ascension of 90.53° and a declination of –66.30° (52000 equinox) or equivalently 128.5° and –89.66° in ecliptic coordinates with a 3.9° margin of error. The surface area is 0.393 square kilometers, the volume is 0.018378 cubic kilometers with a 5% margin of error, and the three axis lengths are 535 meters by 294 meters by 209 meters. The global Itokawa revealed a boomerang-shaped appearance composed of two distinct parts with partly faceted regions and a constricted ring structure.

A telescopic camera on Hayabusa, the Asteroid Multiband Imaging Camera (AMICA) (1), provided about 1400 pictures of Itokawa during its rendezvous phase (2). Members from the AMICA science team and the Guidance and Navigation Control (GNC) team formed a special task force to investigate Itokawa's shape and properties, based on images from AMICA. The task force derived the pole orientation and rotation axis and then formed a global shape model of Itokawa based on a subset of AMICA images. At the beginning of the rendezvous phase, Itokawa was scanned circularly by AMICA. This generated side views, whose mean

interval was about 2°, as well as two polar views. The total number of images was 212, and their resolution ranged from 0.7 to 0.3 m per pixel.

Itokawa's pole orientation was determined by optically tracking feature points, called ground control points (GCPs). The final pole orientation showed retrograde behavior; a right ascension of 90.53° and a celestial declination of –66.30°, or ecliptically as 128.5° and –89.66° with a 3.9° margin of error. The latest ground-based results confirmed the retrograde behavior of Itokawa and its rotation axis, which is almost perpendicular to the ecliptic plane (3–5). Because Itokawa's rotation axis is perpendicular to the ecliptic plane,

AMICA enabled global coverage without polar night regions during a rendezvous that lasted less than half of the orbital time of Itokawa. The north direction (+z) of Itokawa was defined by the International Astronomical Union's right-hand-rule of its own rotation (6), and the +z axis was set to the minimum principal axis of inertia. The direction of the prime meridian (+x) was defined by a GCP, the Black Boulder. The rest of the +y was fixed. This coordinate system was rotated 180° around the z axis with that of previous shape models (4, 7, 8).

The global shape of Itokawa was reconstructed with image-based modeling that integrated limb profiles and stereogrammetric procedures (9). The total number of automatically selected feature points for three-dimensional shape modeling was 308,205, which were related to boulders rather than small distinct craters. Spacing of the points was not ordered, and smooth terrains revealed an absence of such

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boulders. These points were converted to a polygon model with about 3° of resolution (Fig. 1). The total number of polygons was 4285. The model revealed the following shape properties: a surface area of 0.393 km^2 ; a volume of $1.84 \times 10^{-2} \text{ km}^3$ with a 5% margin of error; the size along three axes of 535, 294, and 209 m with a 1-m margin of error; and a bounding box size shown to be $550 \times 298 \times 244 \text{ m}$ with a 1-m margin of error. The spin period of Itokawa was 12.132 hours. The volume precision of Itokawa was controlled by ambiguity of perspective, and those of other parameters were controlled by measuring the lengths on the images. The shape model was formed mainly with stereogrammetry. Previous shapes of Itokawa's envelope are consistent with AMICA images, but detailed features such as the boomerang-shaped bending were hard to identify before the rendezvous. Predictions from previous research with ground-based observations proved to be consistent with this result except for the length of z , the minimum principal axis (5, 7–9), with the most accurate estimations observed by a satellite with a thermal infrared surveyor (10).

One of the most remarkable features of the global shape seems to be two parts, a small "head" with a large "body." These two parts show a boomerang-shaped figure that seems to be separated by a constricted "neck" region, which has a concave ring feature (Figs. 1 and 2). The northern depression of the ring is 20 m in depth and 60 to 120 m in width. The steep slope on the head side of the depression has landslide materials at the bottom (Fig. 2). The southeastern part of the neck is buried with regolithic materials that form part of the Muses Sea, but the steep slope of the head side is continuous to the Shirakami region (Fig. 3). Angles between the long axes of both parts show 24° around the z axis and 26° around the y axis. Itokawa has a considerable concave area, which cannot be projected to the spherical coordinates of geocentric longitude and latitude. The size of the ellipsoid fitted to the body shows $490 \times 310 \times 260 \text{ m}$, and that to the head shows $230 \times 200 \times 180 \text{ m}$ with a 10-m margin of error. A normal vector of the ring's plane to the x axis shows rotations of 7° around the z axis and -20° around the y axis. This constricted ring structure with related features implies that Itokawa may have been formed from the collision of two asteroids (Figs. 2 and 3) (2).

The surface of Itokawa was classified via stereogrammetric observation of images obtained mostly from an altitude of 7 km and a resolution of 0.7 m per pixel [home position (2)]. Stereogrammetrically derived surface textures with reliefs enabled geologic classification and region splitting in Fig. 3, as well as the identification of rough/smooth terrains and facets, large boulders, and major depressed areas. Characteristics of the boulders and rough/smooth terrains are discussed

in (11). Here we primarily discuss the presence and distribution of facets on Itokawa.

The surface of Itokawa appears to have a polyhedral angular structure, with each face of the angular structure called a facet, although their scales ($<100 \text{ m}$) are much smaller than those observed on other asteroids (12). Most facets are relatively flat, although a few have shallow concavity. These facets are divided by gently elevated boundary zones consisting of polyhedron edges (12). Figure 3 shows the boundaries of the facets and large blocks as broken lines. Although the facets are distributed globally on Itokawa, they are more abundant at both ends of the long axis (the head and tail of the body). The most remarkable feature of the head is its completely faceted surface. At the end of the body is the largest facet named "Little Woomera." Although we may include the faceted area as part of the rough terrain, the abundance of large boulders 10 or more meters in diameter showed that the Little Woomera facet has fewer boulders than found on other rough terrains. Traditional facets show the same behavior (13). Some facet boundaries are brighter in reflectance as well (11). The rim of the Little Woomera, which probably resulted from an impact, is one of the most distinct bright zones on Itokawa.

Each polyhedral, angular, and relatively flat structure revealed a shallow concave-like shape. Because this structure was especially distinctive in the shape model, we specifically use the term facet for the polyhedral angular structure, as is often used to describe similar structures on asteroids (Fig. 3). This kind of structure is difficult to recognize only from a single image, mostly because of the randomly-distributed boulders of various sizes. However, circular depressions, most of which would be impact craters, can be identified as specific and isolated patterns. The process that created these facets seem different from that which made the craters. If facets are craters, their sequential order of impacts can be reconstructed by observing the relationship among their emplacements. The appearance of a craterlike and isolated feature, such as the Little Woomera, shows the most recent impact feature. Although the head is entirely covered with facets, all of these facets are distinguishable from obvious craters or circular depressions. On the other hand, all facets adjoining the two smooth terrains, Sagami-hara, and the Muses Sea, are separated abruptly like coast lines, which are boundaries between rough and smooth terrains. This relationship suggests that the smooth terrains may be regolithic extensions over the rough

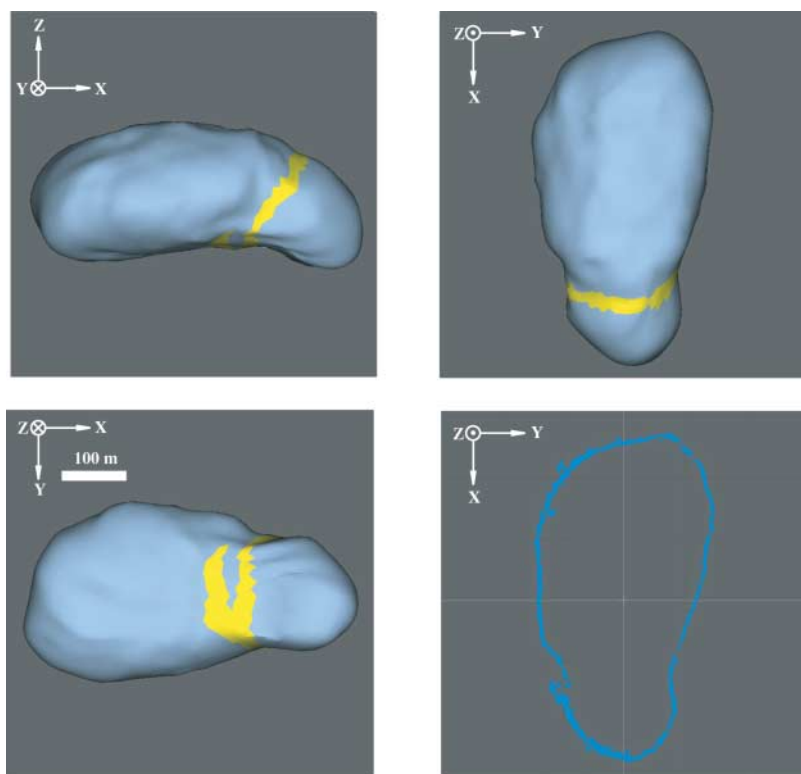


Fig. 1. Three views of Itokawa's shape model and a detailed cross section along the equator. The z direction is north of Itokawa, and the x direction is the prime meridian. The most substantial structure is two pieces (head and body) separated by a circular concave structure (neck). To help guide the eye, a headward slope of the neck is colored in yellow; this region is defined by surface texture of deposits and mass movements derived from the head. A quasi-eyeballed feature of southern polar region shows a local hill (named Yatsugatake in Fig. 3) surrounded by the deposits, but the hill might be an emplacement.

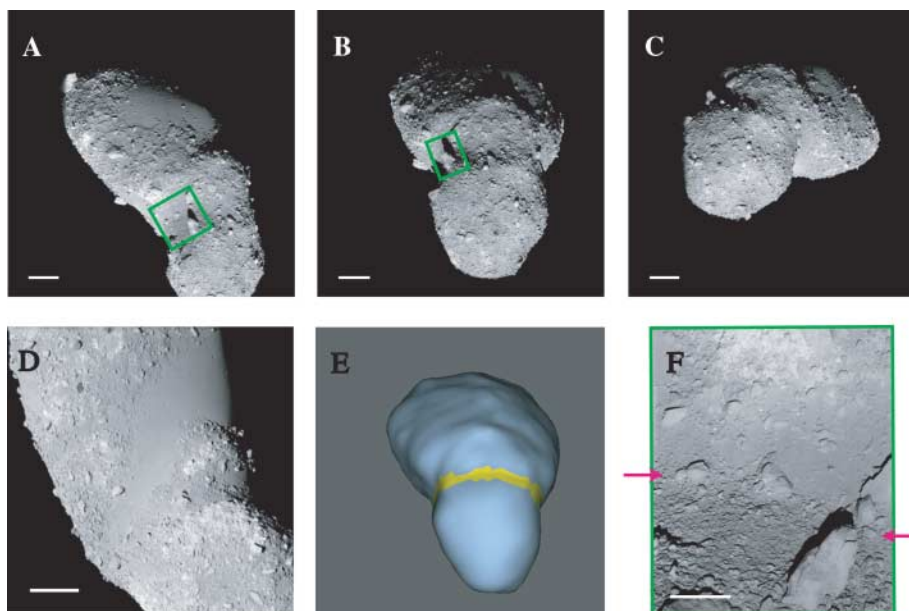


Fig. 2. Views of the neck structure. (A) Northwestern view with image identification code ST_2485860275. (B) Headward north view with ST_2482160259. (C) Northwest view with ST_2485070948. (D) Southeast view with ST_2474846738. (E) Shape model with the same view of the green rectangle in (B). (F) Close-up view (ST_2559003068) of the green rectangle of the neck in (A) and (B). (A) to (C) and (E) show a clear concave ring structure: the neck. (D) shows the southern neck. A brighter local hill is surrounded by gray, rough materials, which underwent a landslide, as shown in yellow in Fig. 1. This hill is called Yatsugatake in Fig. 3. (F) is a close-up of landslides from the head. The red arrow shows its front. Scale bars in (A) to (D), 50 m; in (F), 20 m.

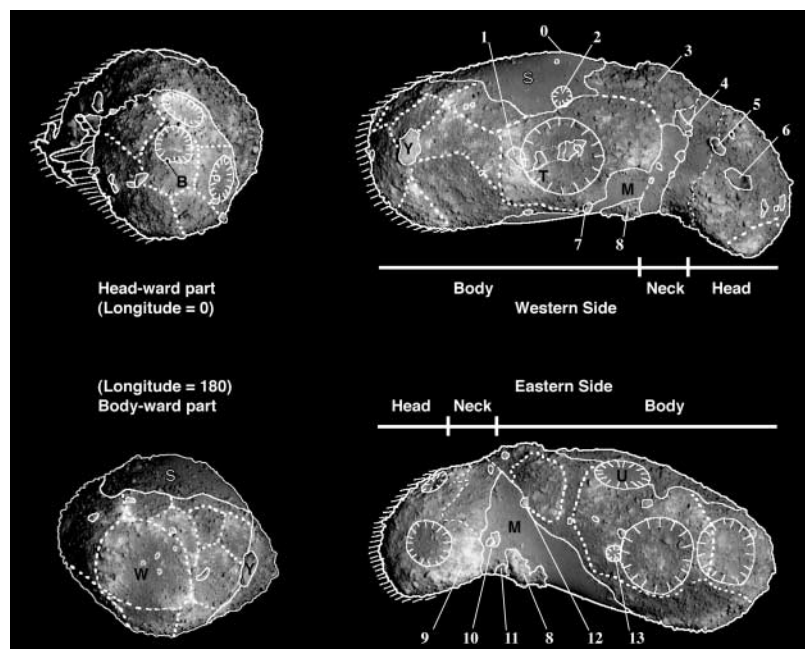


Fig. 3. Facets and nomenclature of Itokawa. Solid lines are boundaries of rough/smooth terrains (11), outlines of Itokawa, and major boulders. Broken lines are boundaries of facets or pedestal blocks. Circular features with spikes are concave features. Hatched regions of left outlines are shadows near the boundaries between day and night. (Image IDs: ST_2423460670, ST_2421011334, ST_2424157005, ST_2417520833). The following IDs show major landmarks: B, the Black Boulder as the landmark of the prime meridian; Y, Yoshinodai boulder; T, Tsukuba Region; M, Muses Sea Planitia; S, Sagami-hara Planitia; W, Little Woomera Region; U, Uchinoura Region; O, the north vertex (+z pole); 1, Kakuta boulder; 2, Fuchinobe crater; 3, Sanriku ridge; 4, Kokubunji boulder; 5, Pencil boulder; 6, M-V boulder; 7, Hilo boulder; 8, Yatsugatake ridge; 9, Shirakami slope; 10, Mountainview boulders; 11, Noshiro smooth terrain; 12, Usuda boulder; and 13, Komaba crater.

terrains. Some facets show gaps at their boundaries, which are different from other plane polygons. These pedestal facets or “blocks” are found on both sides of the body. These appear to be a large pile of rubble. Apparently, Itokawa experienced about 20 or more substantial impacts, which pockmarked the surface. It is not certain whether the facet formation took place before or after the collision of the two larger elements, which currently comprise the head and the body. The neck zone of Itokawa seemingly lacks a faceted structure, which may have been erased at the time of collision between the head and the body. However, the local steep slope, as well as the transported regolithic materials, probably modified the surface structure of the neck zone recently.

Some implied impact structures are found, but little evidence of a global system of groove structures is observed on Itokawa. Although other suspicious short grooves are observed, these are faint features that are hardly separated from the sequential patterns of the boulders. We observed no long, distinct grooves or lineaments on Itokawa like those on Eros (14). This may imply that Itokawa is not a globally coherent body. However, the layer of scattered boulders on the surface might have obscured the presence of underlying grooves or lineaments.

Itokawa has revealed itself as a boomerang-shaped asteroid, covered with many polygonal facets. Formed by the collision of two large masses, Itokawa not only rotates more slowly than other subkilometer asteroids, but it may also be composed of rubble held together by gravity instead of being one monolithic whole. To better understand Itokawa, future research will require investigation of the facets and craters to clarify Itokawa’s collisional history, as well as investigation of the potential existence of hidden grooves to reveal more about Itokawa’s interior structure.

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